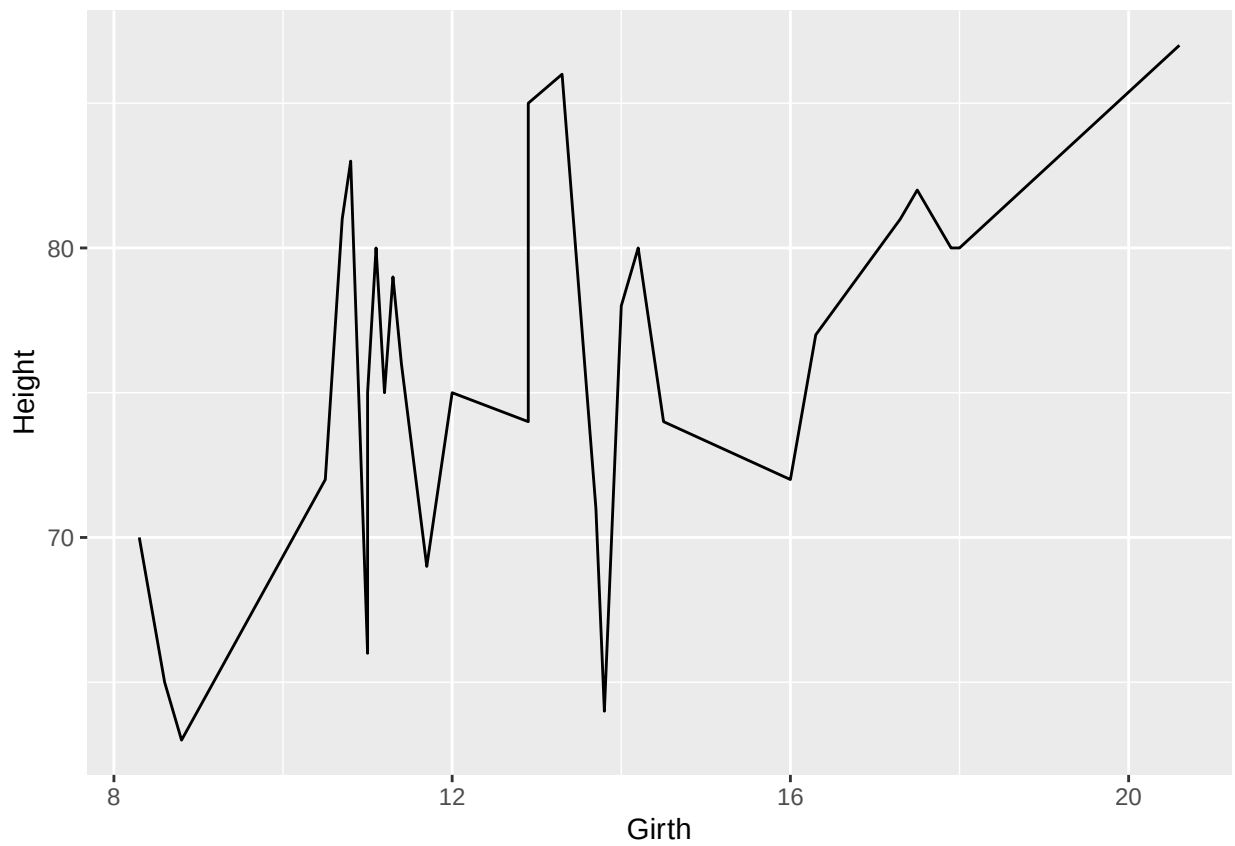


Line Charts

Noratiqah Mohd Ariff

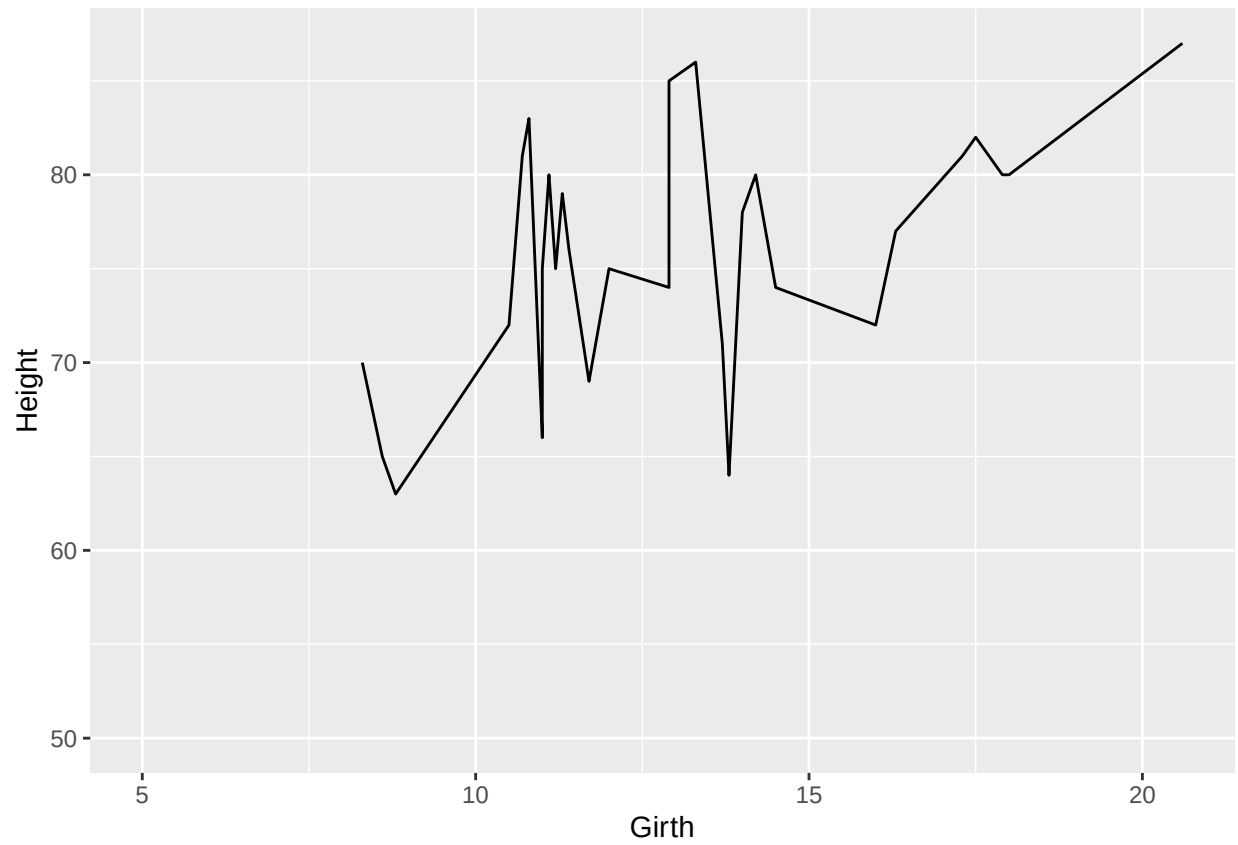
Basic Line Graph

```
g<-ggplot(trees, aes(x=Girth,y=Height))  
g+geom_line()
```

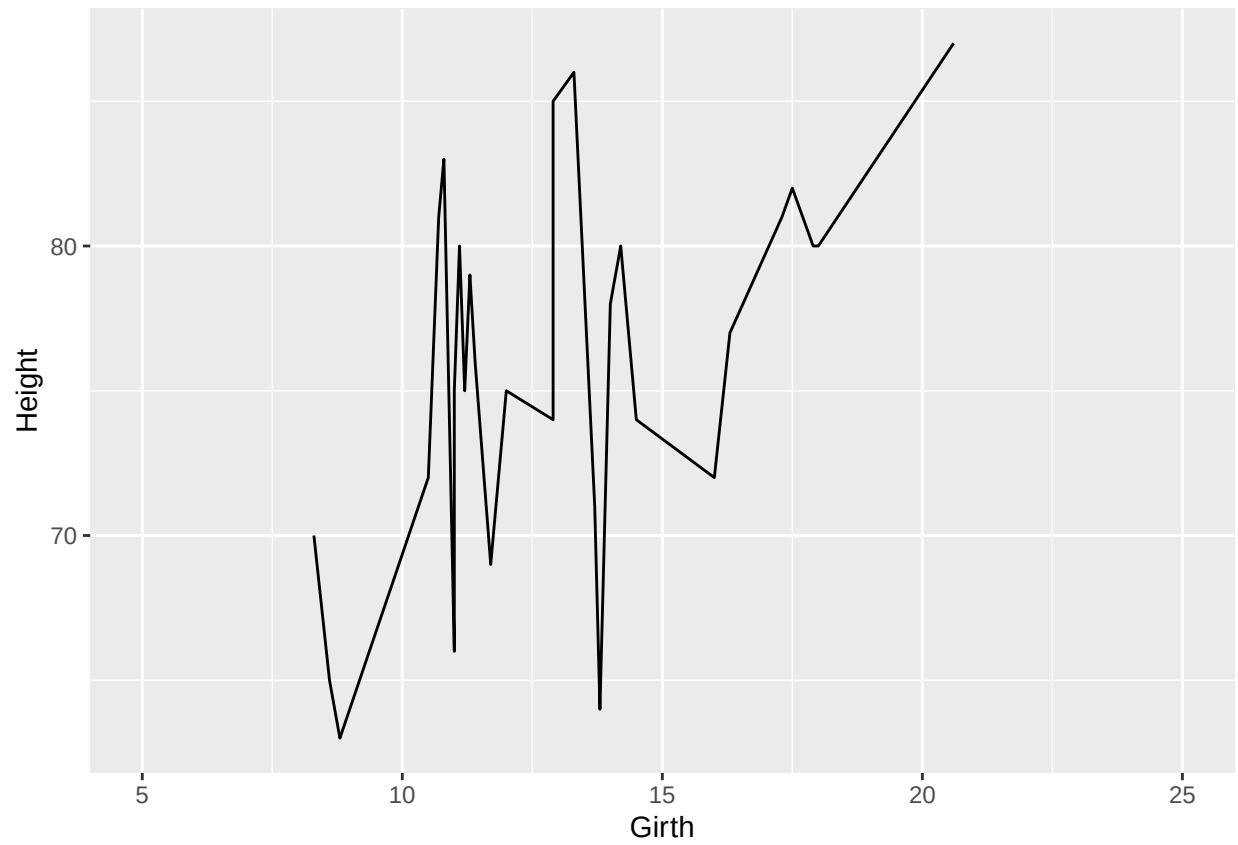


Expand the Limits

```
g<-ggplot(trees, aes(x=Girth,y=Height))  
g+geom_line()+expand_limits(x=5,y=50)
```

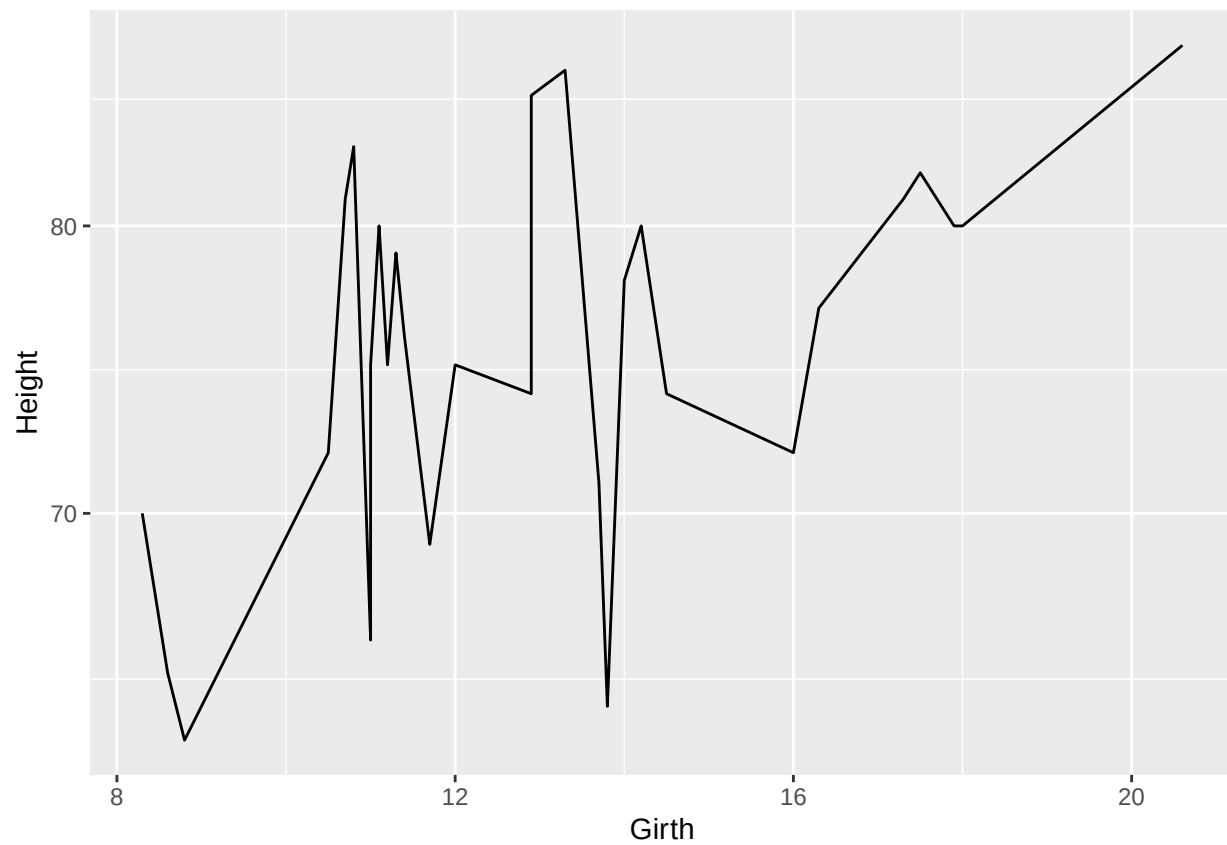


```
g+geom_line()+expand_limits(x=c(5,25))
```

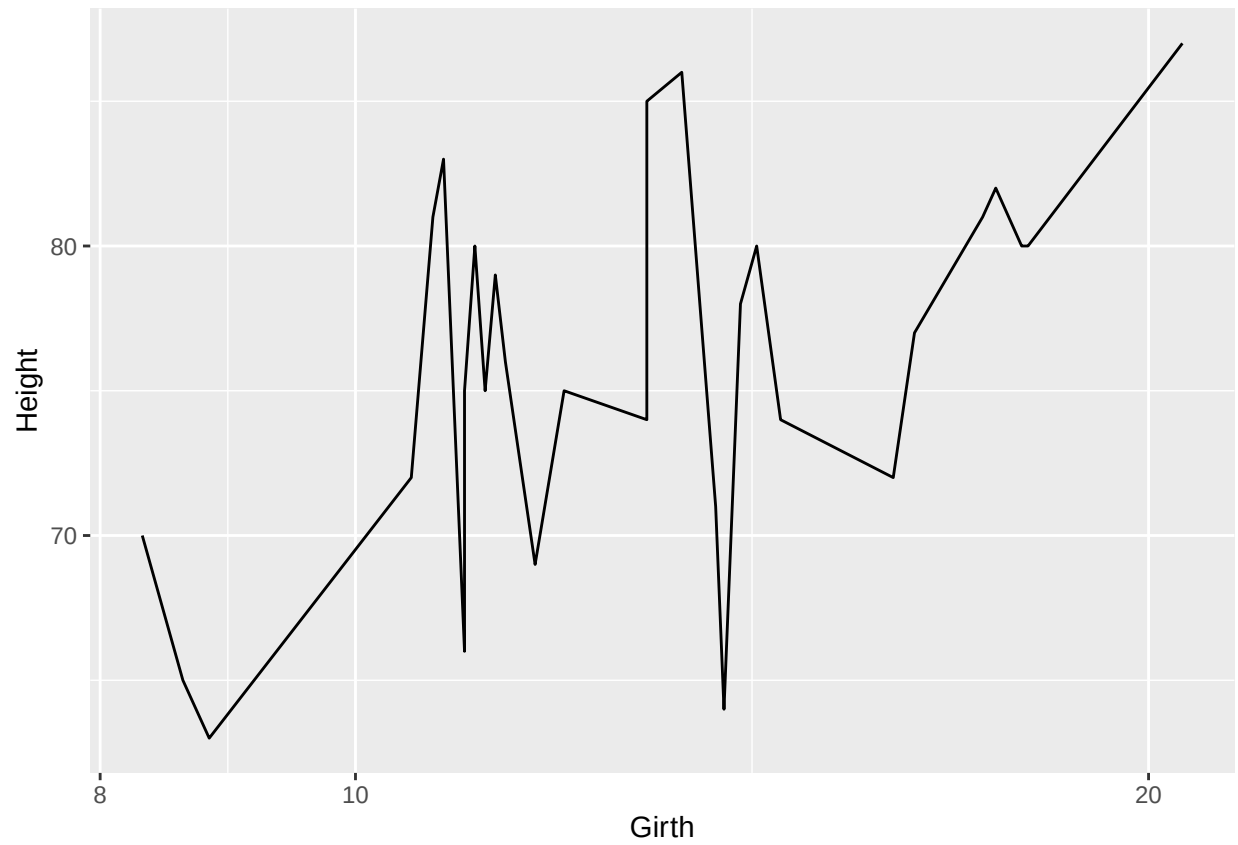


Log-Scale Limits

```
g<-ggplot(trees, aes(x=Girth,y=Height))  
g+geom_line()+scale_y_log10()
```

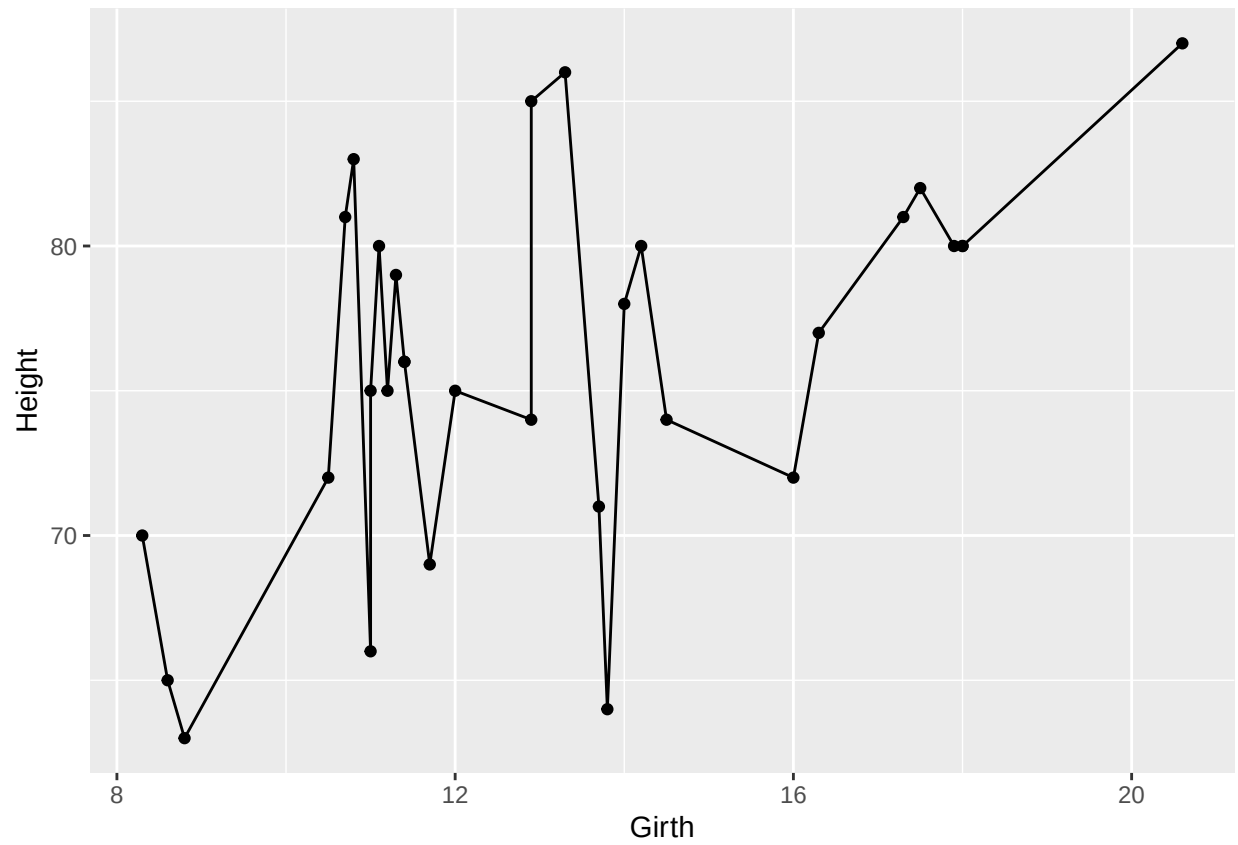


```
g+geom_line()+scale_x_log10()
```



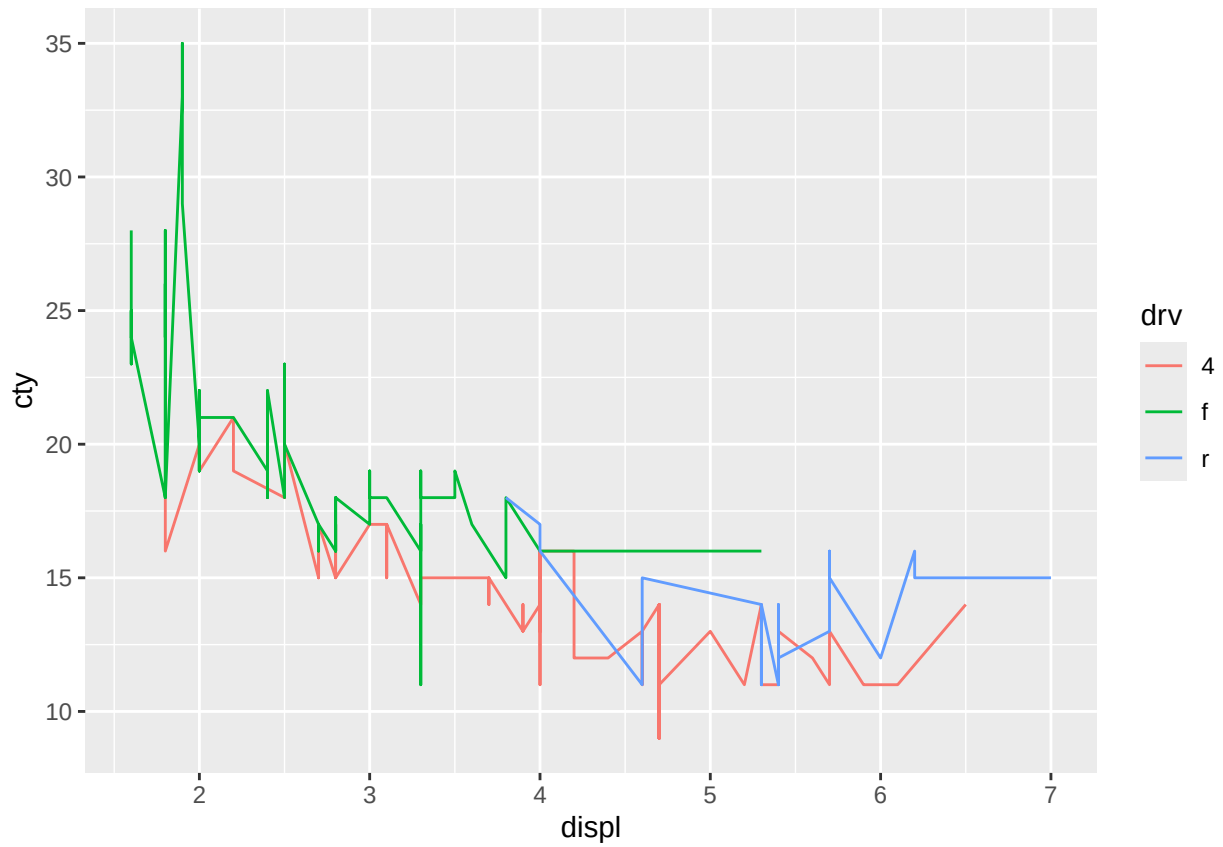
Add Points to A Line

```
g<-ggplot(trees, aes(x=Girth,y=Height))  
g+geom_line()+geom_point()
```

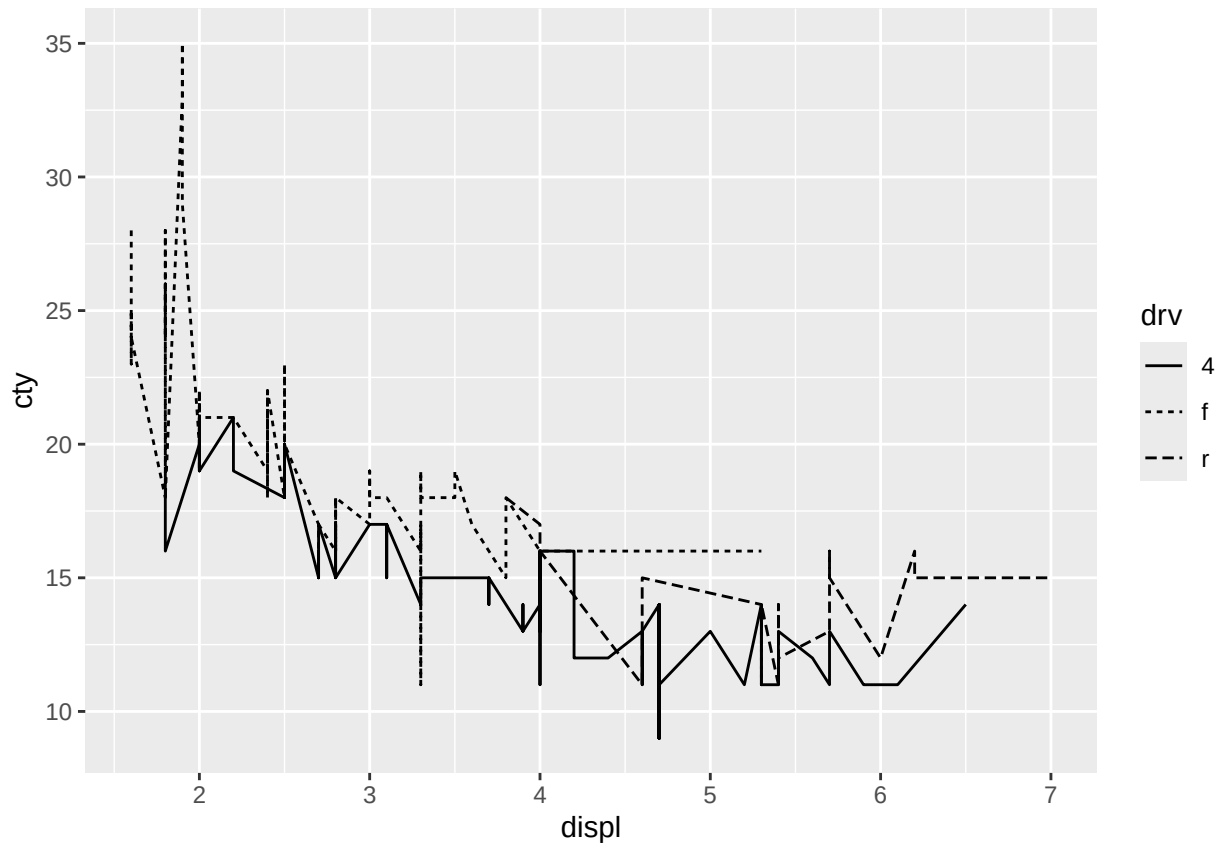


Multiple Lines

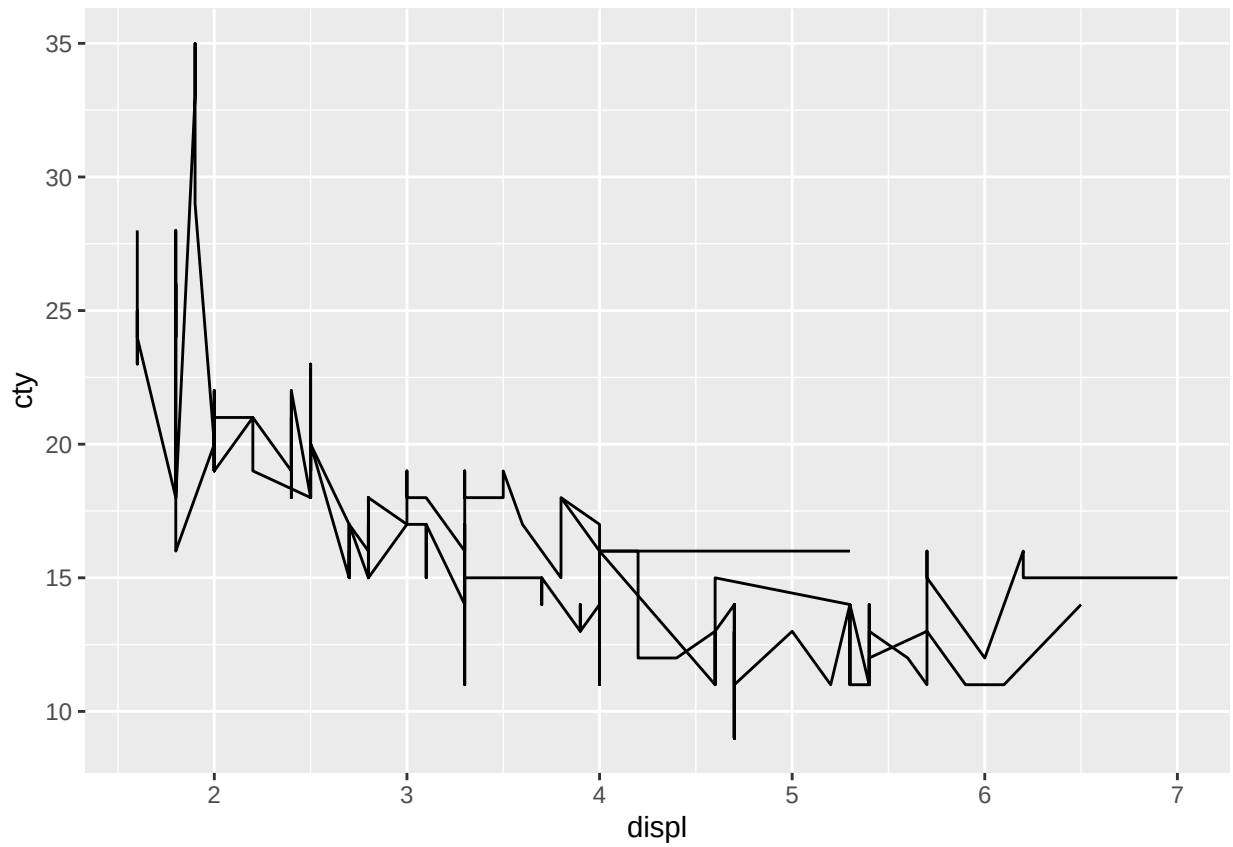
```
ggplot(mpg,aes(x=displ,y=cty,colour=drv))+geom_line()
```



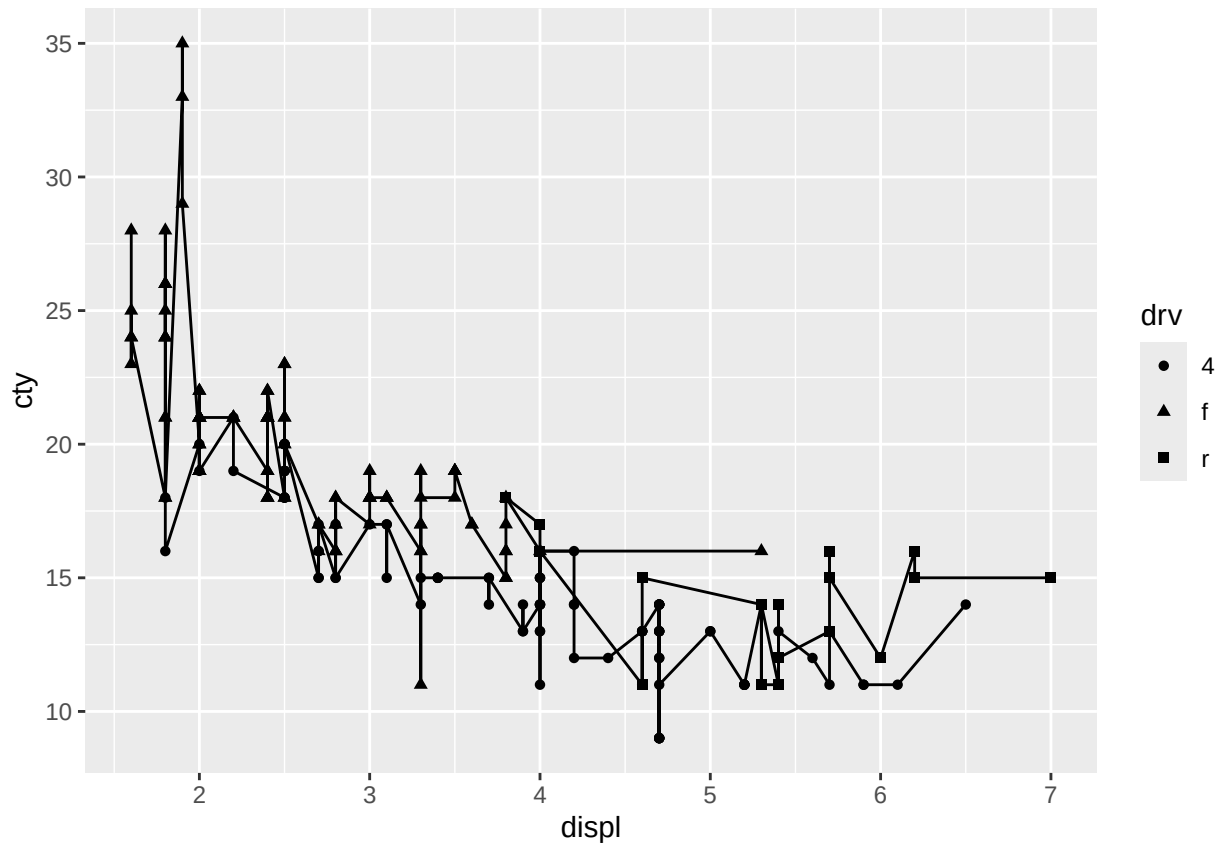
```
ggplot(mpg,aes(x=displ,y=cty,linetype=drv))+geom_line()
```



```
ggplot(mpg, aes(x=displ, y=cty, group=drv)) + geom_line()
```

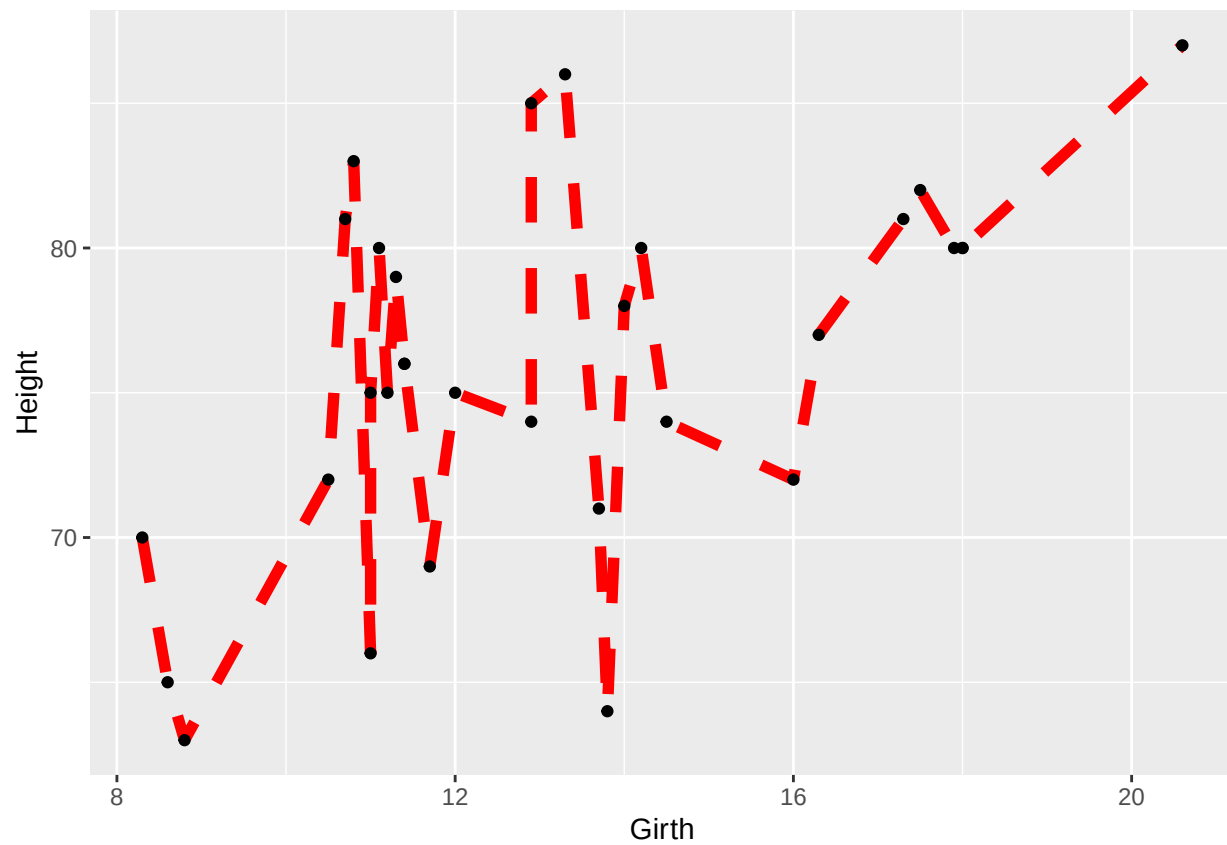



```
ggplot(mpg,aes(x=displ,y=cty,shape=drv))+geom_line()+geom_point()
```

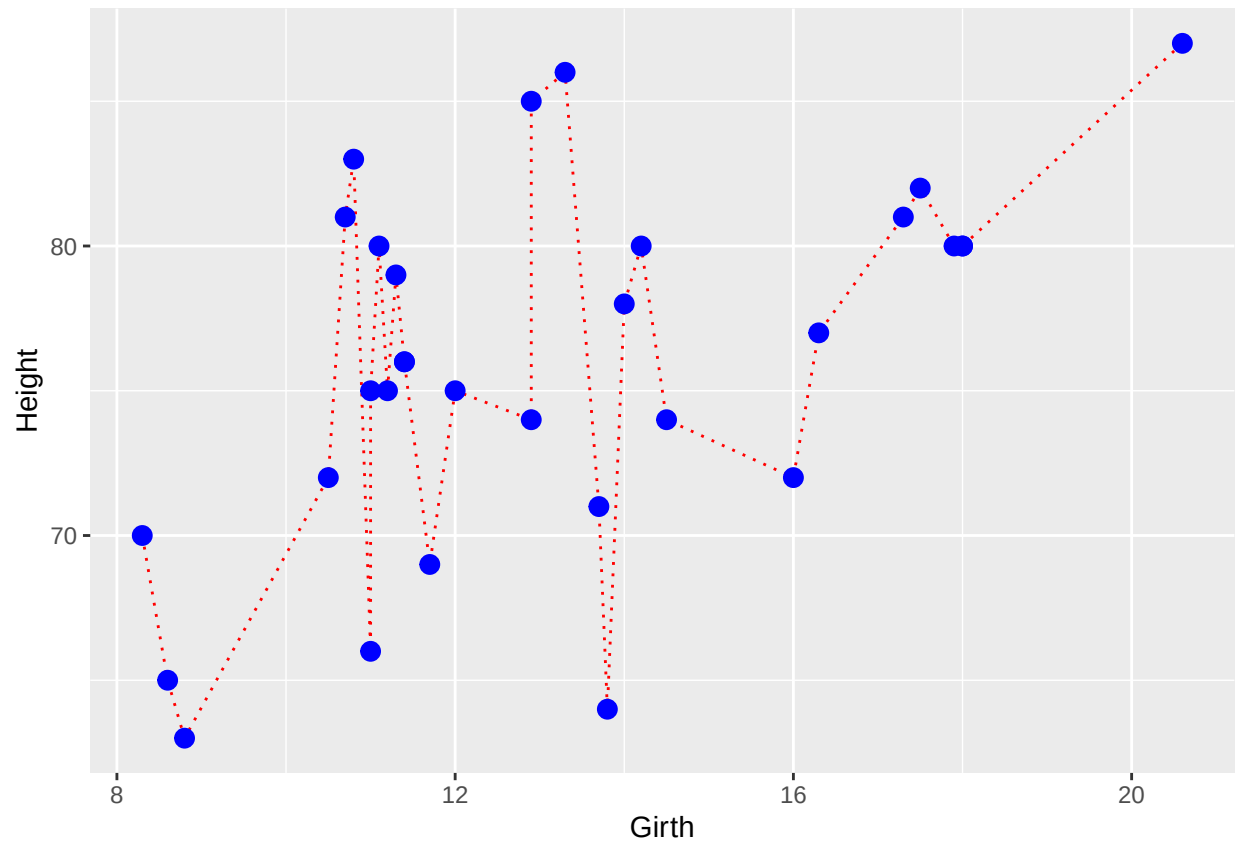


Change Line and Point Features

```
g<-ggplot(trees, aes(x=Girth,y=Height))
g+geom_line(colour="red",linetype="dashed",size=2)+geom_point()
```

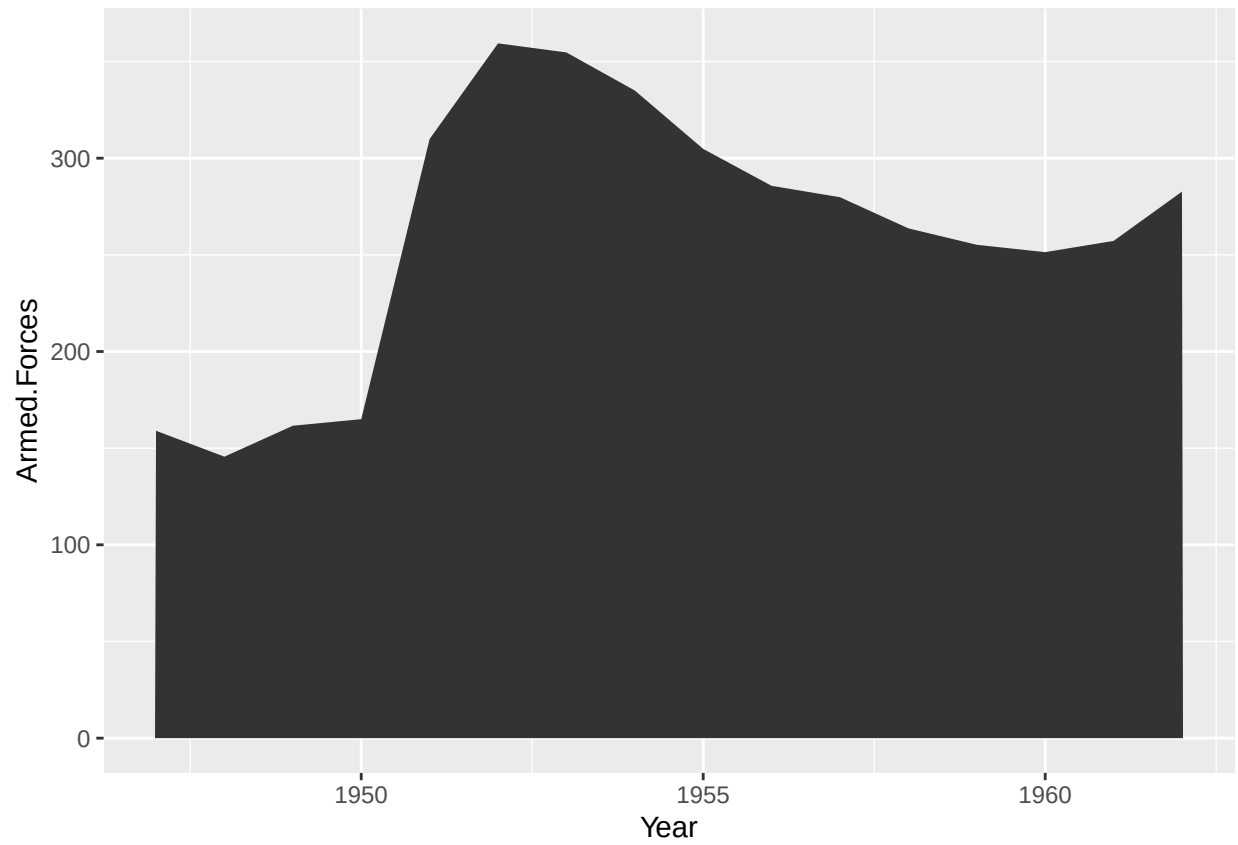


```
g+geom_line(colour="red",linetype="dotted",size=0.5)+
geom_point(shape=19,size=3,colour="blue")
```



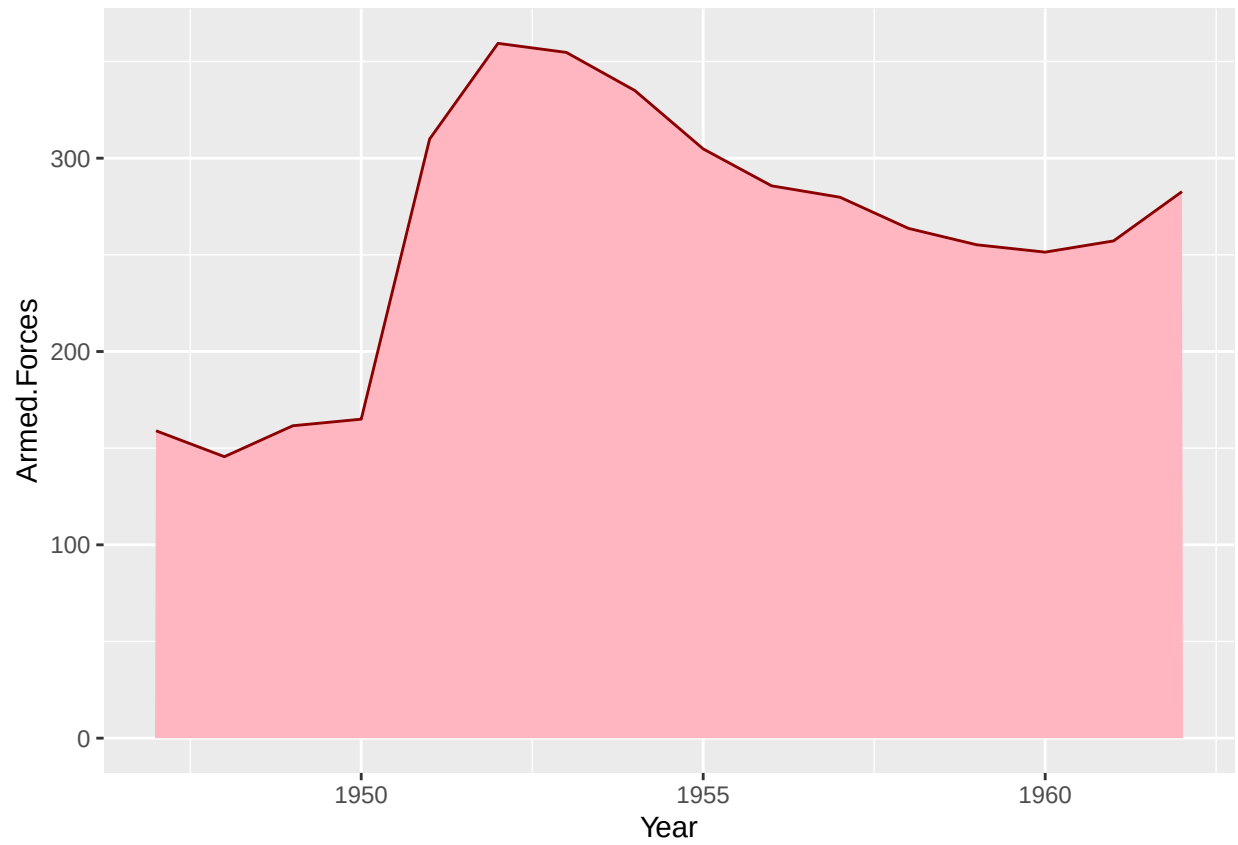
Area Graph

```
ga<-ggplot(longley,aes(x=Year,y=Armed.Forces))  
ga+geom_area()
```

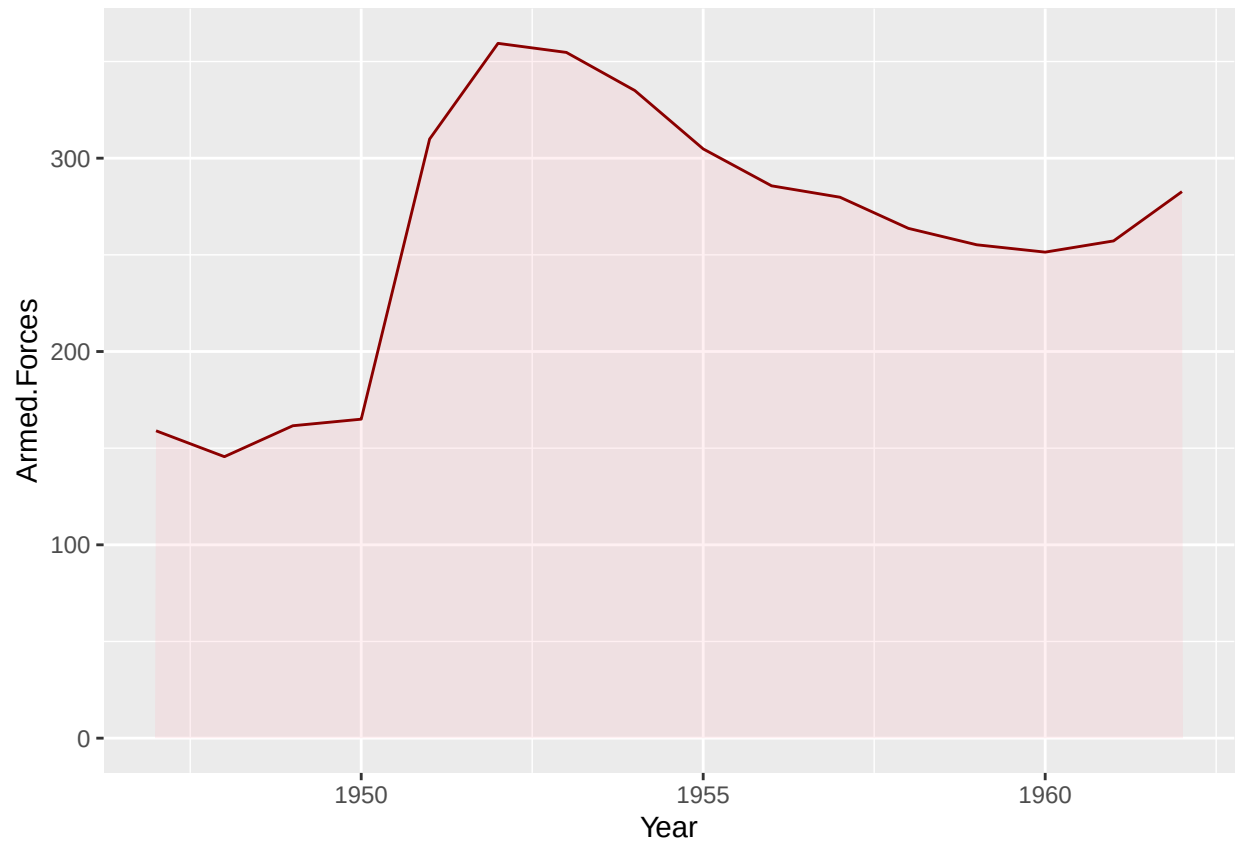


Change Fill

```
ga<-ggplot(longley,aes(x=Year,y=Armed.Forces))  
ga+geom_area(fill="lightpink",colour="darkred")
```

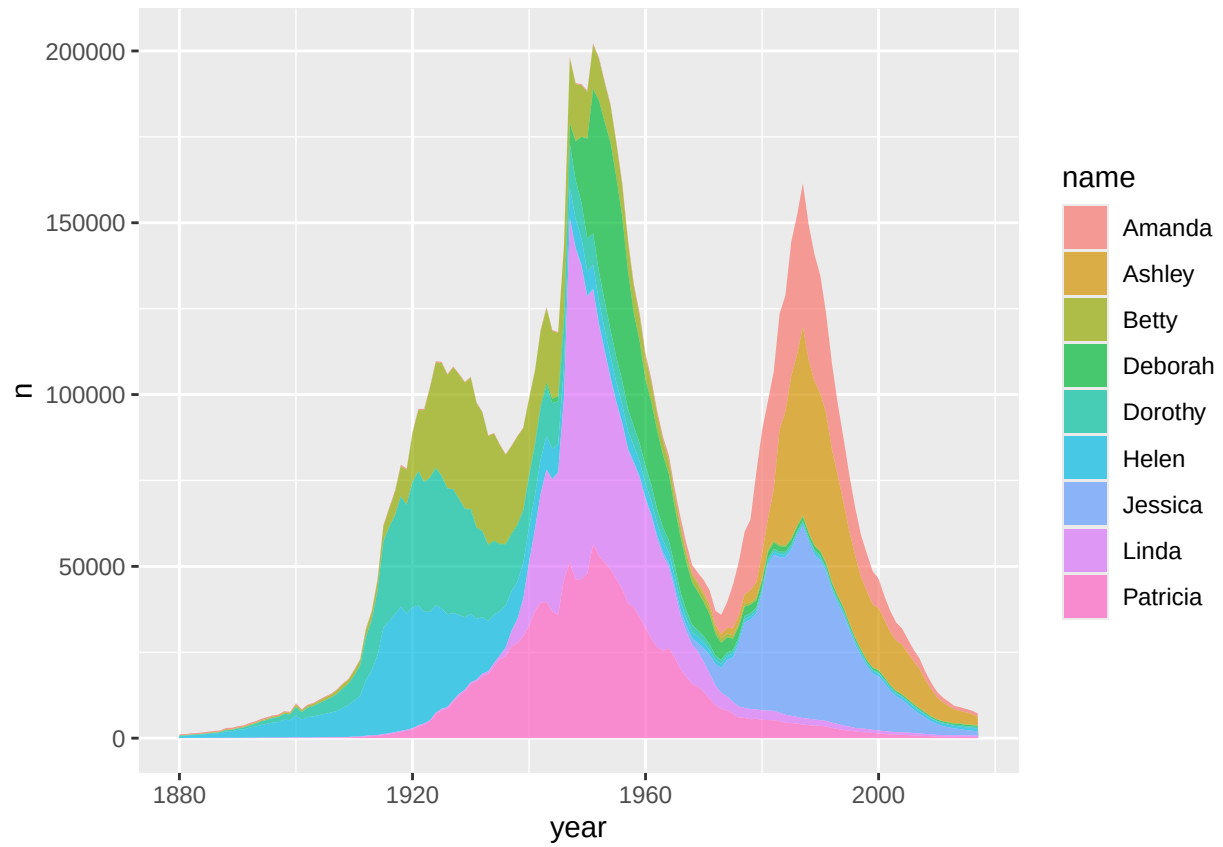


```
ga+geom_area(fill="lightpink",colour="darkred",alpha=0.2)
```



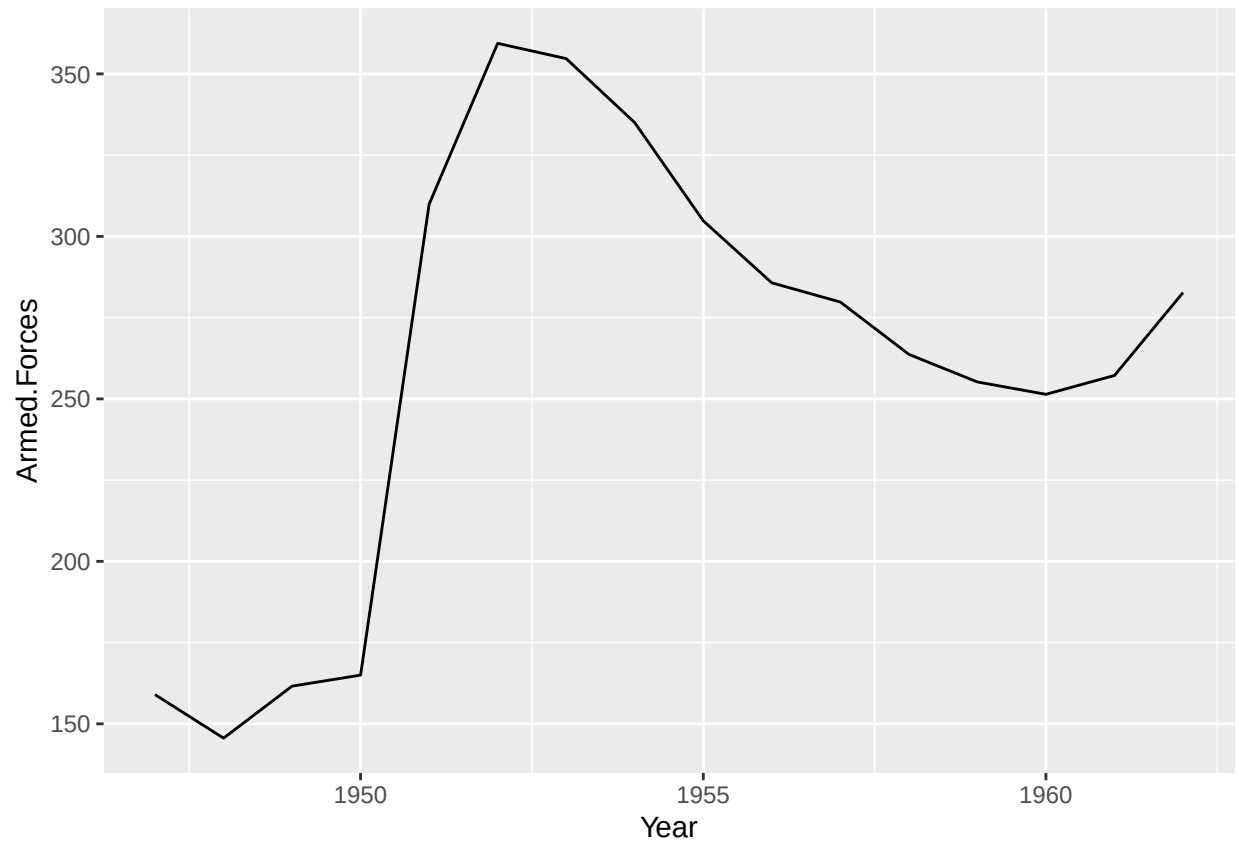
Stacked Area Graph

```
library(babynames)
baby<-subset(babynames,subset=(sex=="F"&
name %in% c("Ashley","Amanda", "Jessica","Patricia","Linda","Deborah","Dorothy","Betty","Helen")))
gs<-ggplot(baby,aes(x=year,y=n,fill=name))
gs+geom_area(alpha=0.7)
```



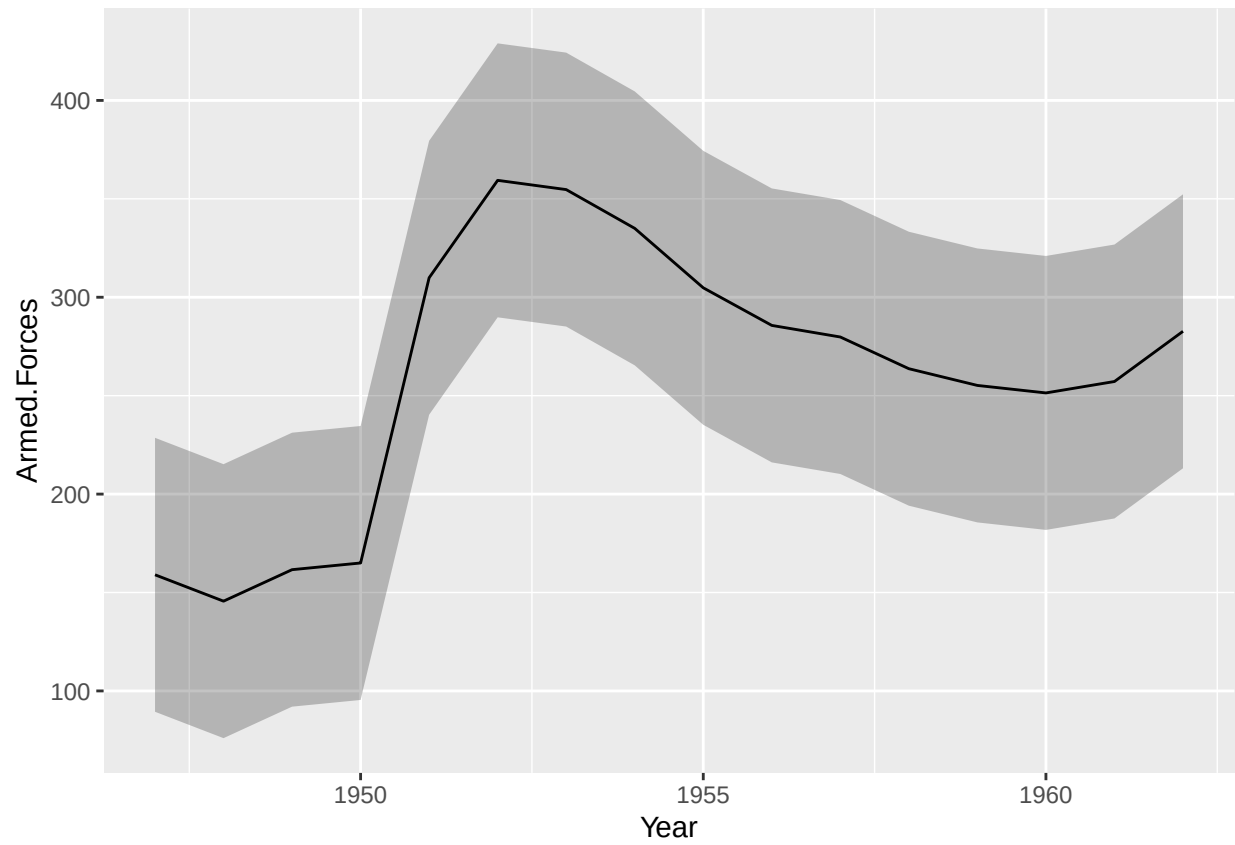
Add Confidence Intervals

```
ga+geom_line()
```

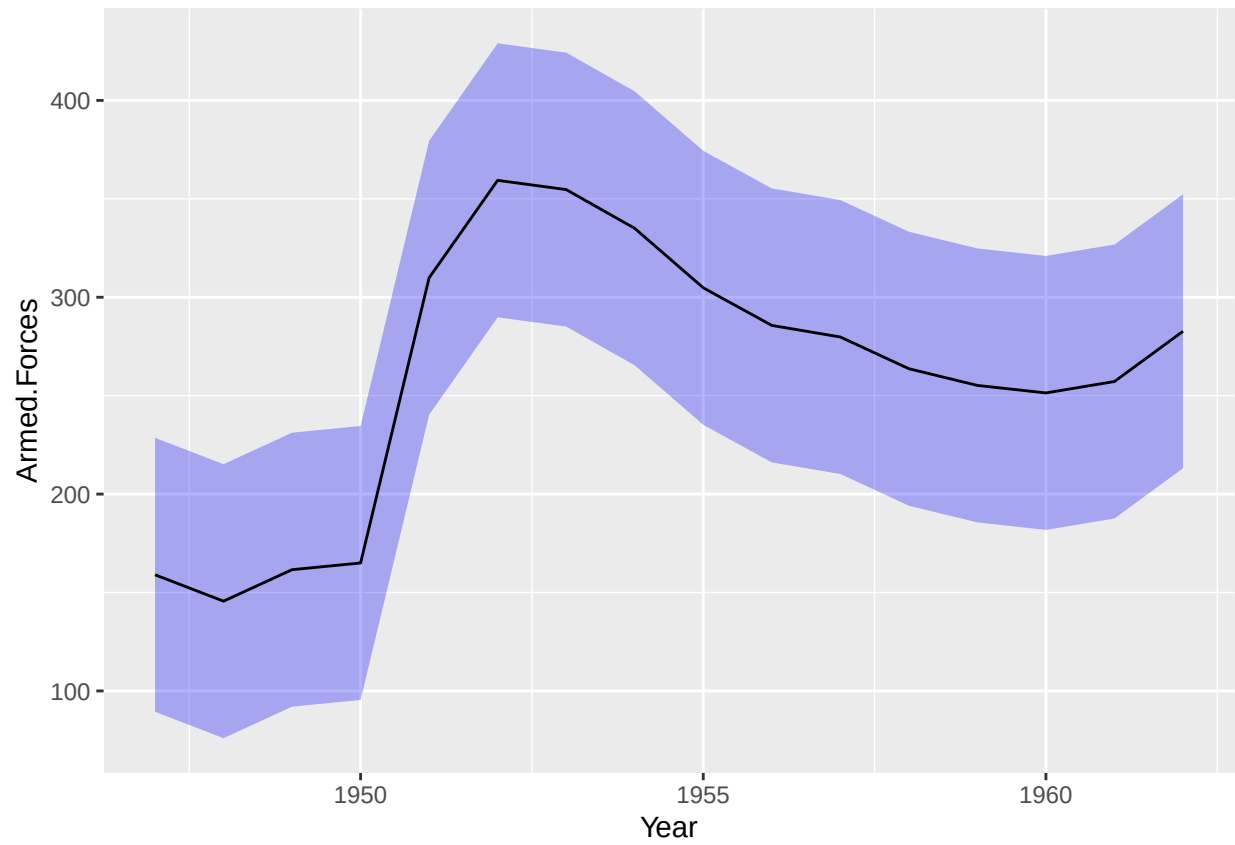



```
std<-sd(longley$Armed.Forces)

ga+geom_ribbon(aes(ymin=Armed.Forces-std,ymax=Armed.Forces+std),alpha=0.3)+
geom_line()
```



```
ga+geom_ribbon(aes(ymin=Armed.Forces-std,ymax=Armed.Forces+std),alpha=0.3,fill="blue")+  
geom_line()
```



```
ga+geom_line(aes(y=Armed.Forces-std),linetype="dashed",colour="darkgrey")+  
geom_line(aes(y=Armed.Forces+std),linetype="dashed",colour="darkgrey")+  
geom_line()
```

